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Course: Calculus Beta

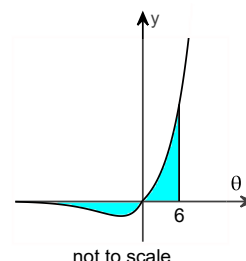
Assignment: MyLab Math - Uge 3

The integral in this exercise converges. Evaluate the integral without using a table.

$$\int_{-\infty}^6 \theta e^{\theta} d\theta$$

Integrals with infinite limits of integration are improper integrals of type I. Integrals of functions that become infinite at a point within the interval of integration are improper integrals of type II.

The function $y = \theta e^{\theta}$ is graphed to the right. Notice that $\int_{-\infty}^6 \theta e^{\theta} d\theta$ is an improper integral of type I.



If $f(x)$ is continuous on $(-\infty, b]$, then $\int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) dx$. Because θe^{θ} is continuous on $(-\infty, 6]$, use this property to rewrite

$$\int_{-\infty}^6 \theta e^{\theta} d\theta.$$

$$\int_{-\infty}^6 \theta e^{\theta} d\theta = \lim_{a \rightarrow -\infty} \int_a^6 \theta e^{\theta} d\theta$$

To integrate, use integration by parts. The integration by parts formula for definite integrals is shown below.

$$\int_a^b u(\theta) v'(\theta) d\theta = u(\theta) v(\theta) \Big|_a^b - \int_a^b v(\theta) u'(\theta) d\theta$$

For $\int_a^6 \theta e^{\theta} d\theta$, assign u and dv so that $u(\theta) v(\theta) \Big|_a^b - \int_a^b v(\theta) u'(\theta) d\theta$ can be evaluated more easily than $\int_a^b u(\theta) v'(\theta) d\theta$. To do so, make sure that $v(\theta) u'(\theta) d\theta$ is easily integrable. Let $u = \theta$ and let $dv = e^{\theta} d\theta$.

Find du and v .

$$\begin{aligned} u &= \theta & dv &= e^{\theta} d\theta \\ du &= d\theta & v &= e^{\theta} \end{aligned}$$

Use the integration by parts formula $\int_a^b u(\theta) v'(\theta) d\theta = u(\theta) v(\theta) \Big|_a^b - \int_a^b v(\theta) u'(\theta) d\theta$ to rewrite $\int_a^6 \theta e^{\theta} d\theta$.

$$\begin{aligned} \int_a^6 \theta e^{\theta} d\theta &= \theta e^{\theta} \Big|_a^6 - \int_a^6 e^{\theta} d\theta \\ &= 6e^6 - ae^a - \int_a^6 e^{\theta} d\theta \end{aligned} \quad \text{Substitute the limits of integration.}$$

Evaluate $\int_a^6 e^{\theta} d\theta$.

$$\begin{aligned} 6e^6 - ae^a - \int_a^6 e^{\theta} d\theta &= 6e^6 - ae^a - [e^{\theta}]_a^6 \\ &= 6e^6 - ae^a - (e^6 - e^a) \end{aligned} \quad \text{Substitute the limits.}$$

Now find the limit of this expression as a approaches $-\infty$.

$$\lim_{a \rightarrow -\infty} (6e^6 - ae^a - e^6 + e^a)$$

Begin by finding $\lim_{a \rightarrow -\infty} ae^a$. Notice that attempting to substitute $-\infty$ for a would result in $\frac{-\infty}{e^{\infty}}$, or $\frac{-\infty}{\infty}$. Instead, try another method.

Write ae^a as $\frac{a}{e^{-a}}$. Since $\lim_{a \rightarrow -\infty} a = -\infty$ and $\lim_{a \rightarrow -\infty} e^{-a} = \infty$, apply L'Hôpital's Rule. Differentiate the numerator and denominator with respect to a .

$$\lim_{a \rightarrow -\infty} \left(\frac{a}{e^{-a}} \right) = \lim_{a \rightarrow -\infty} \left(\frac{1}{-e^{-a}} \right)$$

Evaluate this limit.

$$\lim_{a \rightarrow -\infty} \left(\frac{1}{-e^{-a}} \right) = 0$$

Now find $\lim_{a \rightarrow -\infty} e^a$.

$$\lim_{a \rightarrow -\infty} e^a = 0$$

Use these results to find $\lim_{a \rightarrow -\infty} (6e^6 - ae^a - e^6 + e^a)$.

$$\begin{aligned} \lim_{a \rightarrow -\infty} (6e^6 - ae^a - e^6 + e^a) &= 6e^6 - 0 - e^6 + 0 \\ &= 5e^6 \end{aligned}$$

Simplify.

Thus, $\int_{-\infty}^6 \theta e^{\theta} d\theta = 5e^6$.